COPYRIGHT RESERVED BCA (Sem-I) – Programming

Fundamental Using C++ (101) Core-I

2024

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section–A

Objective type Questions

Compulsory

1. Choose the correct answer of the following multiple choice questions : 2×10=20

(i) Which of the following is a correct statement about C++?

a) C++ is a purely procedural language.

b) C++ does not support function overloading.

c) C++ supports both procedural and object-oriented programming.

d) C++ does not use macros.

(ii) What is the correct syntax for an if-else statement?

a) if (condition) { ... } else { ... }

b) if (condition) { ... } else if { ... } else { ... }

c) if (condition) { ... } else { ... }

d) if condition then { ... } else { ... }

(iii) What is the output of the code below?

int $x = 2$;

switch (x)

{case 1: printf("One");

case 2: printf("Two");

case 3: printf("Three");

default: printf("Default"); }

a) Two

b) Two Three Default

c) Default

d) Compilation Error

(iv) Which of the following is not a loop control structure in C++?

a) for

b) while

c) loop

d) do-while

(v) What is the output?

int $x = 10$, $y = 20$;

$x == 20$ && $y != 10$? printf("True") :

printf("False");

a) True

b) False

c) Garbage

d) None of the above

(vi) What is the default return type of a function if not specified?

a) void

b) int

c) float

d) char

(vii) What does #include<iostream.h> do in a C++ program?

a) Defines the structure

b) Links header files to source code

c) Starts execution of the program

d) Imports standard input/output library

(viii) Which of the following supports runtime polymorphism?

a) Function overloading

b) Constructor overloading

c) Virtual functions

d) Inline functions

(ix) What is the scope resolution operator in C++?

a) :

b) ::

c) #

d) ->

(x) Which function is used to read a single character from the user in C?

a) getchar( )

b) scanf( )

c) getch( )

d) cin.get( )

Section–B

Short Answer type Questions

Answer any **four** questions of the following : 5×4=20

2. Explain the difference between procedural and object-oriented programming with examples.
3. Write a swapping program to demonstrate the use of call by reference.
4. Differentiate between structure and union in C++ with examples.
5. Write a program to input a string and check if it is a palindrome or not.
6. What is inheritance? Explain multilevel inheritance with a suitable code.

7. Write a short note on function overloading with example.
8. What are preprocessor directives? Explain the use of macro with an example.

Section–C

Long Answer type Questions

Answer any **two** questions of the following : 15×2=30

9. Write a program to create a class Student with data members name, roll, and marks. Use a constructor to initialize the details. Also demonstrate the use of copy constructor by creating another object. Use a separate display function which must not be an inline function.
10. Write a menu-driven program to perform matrix operations (Addition, Subtraction and Multiplication) using 2D arrays.

11. Write a program using class and inheritance to store and display information about teachers and students. Use runtime polymorphism to achieve function overriding.

12. Explain exception handling in C++ with syntax and a program to handle division by zero.

13. Write a C++ program to sort an array of 10 elements in ascending and descending order using bubble sort.

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